

Synthetic Hospital CAD Research Unit

12-LEAD ELECTROCARDIOGRAM REPORT

Cardiology Non-Invasive Diagnostic Laboratory

Synthetic sample. Not a real patient.

Patient: SAMPLE, TRIPLE-VESSEL CRITICAL CAD

MRN: SYN-0004

DOB: 1954-01-15 (Age / Sex: 72y / Male)

Report Date: 2026-03-15

Encounter: Ambulatory Outpatient

Ref MD: Dr. S. Synthetic, MD

Technical & Global Measurements

Rhythm: Normal Sinus Rhythm

Heart Rate: 88 bpm

PR Interval: 160 ms

QRS Duration: 92 ms

QT / QTc: 400 / 420 ms

Mean QRS Axis: +45° (Normal)

Morphological & Ischemic Findings

Parameter	Result	Diagnostic Criteria / Notes
Pathological Q wave	Present	Leads V1–V4 / II, III, aVF ($> 0.04s$ or $> 25\%$ R-wave)
ST elevation	Present	J-point displacement ≥ 1.0 mm in limb/precordial
ST depression	Present	Horizontal/downsloping ≥ 0.5 mm 80ms post-J
T-wave inversion	Present	Symmetric inversion ≥ 1.0 mm in contiguous leads
Left ventricular hypertrophy	Present	Sokolow-Lyon voltage index ($SV1 + RV5 \geq 35$ mm)
Poor R-wave progression	Present	Loss of anterior R-wave amplitude ($RV3 \leq 3$ mm)
Bundle branch block	None	Intraventricular conduction delay morphology

Impression & Interpretation

Conclusion: ST elevation suggestive of acute or evolving transmural injury; ST depression consistent with subendocardial ischemia; T-wave inversion reflecting repolarization abnormality; pathological Q wave indicative of prior infarction; left ventricular hypertrophy; poor R-wave progression. Correlate clinically.

Over-read by: Dr. E. Cardiologist, MD (Cardiology)

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